DAY 09 OF 10

NumPy & Pandas for Data

Your first step into data analysis and machine learning

Today's agenda

1. Why NumPy and Pandas
2. NumPy arrays
3. Pandas Series and DataFrame
4. Selecting, filtering and grouping
5. Cleaning data

Python in 10 Days
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Why NumPy and Pandas?

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- NumPy: fast numeric arrays and maths on whole arrays at once
- Pandas: tables (DataFrames) for real-world data like Excel / SQL tables
- Almost every data science and ML workflow starts with them
- Install: pip install numpy pandas matplotlib
- Convention: import numpy as np / import pandas as pd

NumPy arrays

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- np.array creates an array
- Operations apply to every element (vectorisation), no loop needed
- Boolean filtering: a[a > 15]
- shape, reshape, arange, mean, sum, max are everyday tools

Python

```python
import numpy as np

a = np.array([10, 20, 30, 40])
print(a * 2)
print(a.mean())
print(a[a > 15])

m = np.arange(6).reshape(2, 3)
print(m)
print(m.shape)
```

OUTPUT

```
[20 40 60 80]
25.0
[20 30 40]
[[0 1 2]
 [3 4 5]]
(2, 3)
```

Pandas DataFrame

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- A DataFrame is a table with labelled rows and columns
- Create from a dictionary of lists
- Each column is a Series
- head(), info(), describe() give a quick overview

Python

```python
import pandas as pd

df = pd.DataFrame({
  "name":   ["Asha", "Ravi", "Sam", "Nisha"],
  "course": ["SQL", "Python", "Python", "SQL"],
  "marks":  [82, 67, 91, 74],
})
print(df.head())
print(df["marks"].mean())
```

OUTPUT

```
    name  course  marks
0   Asha     SQL     82
1   Ravi  Python     67
2    Sam  Python     91
3  Nisha     SQL     74
78.5
```

Selecting and filtering

- df["col"] selects a column; df[["a", "b"]] selects several
- df[condition] filters rows
- Combine conditions with & and | and wrap each in ( )
- loc selects by label, iloc by position
- sort_values orders rows

Python

```python
print(df[df["marks"] > 75])
print(df.sort_values("marks", ascending=False).head(2))
```

OUTPUT

```
   name  course  marks
0  Asha     SQL     82
2   Sam  Python     91
   name  course  marks
2   Sam  Python     91
0  Asha     SQL     82
```

Grouping and summary

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- groupby splits data by a column, then aggregates
- agg lets you apply several functions at once
- value_counts counts occurrences
- Same idea as SQL GROUP BY, which you will see in the SQL course

Python

```python
print(df.groupby("course")["marks"].mean())
print(df["course"].value_counts())
print(df.groupby("course")["marks"].agg(["min", "max", "mean"]))
```

OUTPUT

```
course
Python    79.0
SQL       78.0
Name: marks, dtype: float64
```

Reading files and cleaning data

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- pd.read_csv loads a file; to_csv saves
- isna().sum() shows missing values per column
- fillna fills gaps; dropna removes rows
- drop_duplicates, astype and rename tidy your data
- Cleaning is usually 70% of real data work

Python

```
df = pd.read_csv("students.csv")
df.info()
print(df.describe())
print(df.isna().sum())

df["marks"] = df["marks"].fillna(df["marks"].mean())
df = df.dropna(subset=["name"])
df = df.drop_duplicates()
df.to_csv("clean.csv", index=False)
```

A first chart

- Matplotlib draws charts; Pandas can call it directly
- A chart often explains data faster than a table
- Always add a title and axis labels

Python

```python
import matplotlib.pyplot as plt

df.groupby("course")["marks"].mean().plot(kind="bar")
plt.title("Average marks by course")
plt.ylabel("Marks")
plt.show()
```

Practice time

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1. Create a NumPy array of 1 to 20 and print only multiples of 3
2. Build a DataFrame of 6 products with price and quantity; add a total column
3. Find the top 3 products by total
4. Group sales by city and print the sum for each city
5. Download any small CSV (for example from Kaggle), load it and run info() and describe()

Challenge: find and fix missing values in a dataset of your choice

Day 9 recap

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- NumPy = fast arrays; Pandas = tables
- Select, filter, sort, group, aggregate
- Clean missing and duplicate data before analysis
- This is the base for Machine Learning and Data Science

Homework: finish practice. Next: Mini project and next steps